After second impact, rest; springs are natural.
At their equilibria, rest and move
right together at , uniformly, without further
oscillations or collisions.

For

Since ,

For , after first impact,
and give the table-frame positions and velocities.

If and meet in their later motion, then

From ,

Since ,

The roots are

The least positive nonzero solution satisfies

and

center-frame velocities at meeting, by , :

at